

MOMBASA INTL, KenyaWMO: **638200**Lat: **4.035S**Lon: **39.594E**Elev: **200**StdP: **14.59**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	68.8	69.7	63.8	89.2	80.5	64.9	92.8	80.0	23.2	81.0	21.4	80.3	4.3	210	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	13.3	91.7	77.5	91.1	77.4	89.9	77.1	79.9	86.6	79.2	85.7	79.0	85.4	12.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	147.8	82.8	77.4	143.5	82.1	77.2	142.6	81.9	43.5	87.0	43.0	86.5	42.5	86.3	92.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	18.5	16.7	DB	63.8	95.9	6.3	2.3	59.3	97.5	55.7	98.9	52.1	100.1	47.6	101.8	
			WB	62.0	83.1	5.6	2.8	58.0	85.2	54.7	86.8	51.5	88.4	47.4	90.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.2	82.6	83.1	83.9	82.5	79.6	77.9	76.4	76.3	77.5	79.4	80.9	82.4	(6)
	DBStd	3.07	1.35	1.43	1.44	2.07	1.99	1.47	1.35	1.79	1.35	1.52	1.80	1.43	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11022	1011	928	1051	976	918	836	819	815	826	911	927	1004	(10)
	CDD65	5547	546	508	586	526	453	386	354	350	376	446	477	539	(11)
	CDH74	50313	5517	5407	6847	5689	3689	2595	2009	2030	2573	3728	4659	5570	(12)
	CDH80	15758	2057	2098	2779	2025	932	509	296	292	495	901	1376	1998	(13)
(14) Wind	WSAvg	7.6	8.3	7.6	6.7	7.9	8.7	8.5	8.3	7.8	7.8	7.2	5.7	6.3	(14)
(15) Precipitation	PrecAvg	42.7	1.1	0.6	1.9	6.0	9.5	4.3	2.8	2.6	2.2	4.6	4.3	2.5	(15)
	PrecMax	86.7	8.8	3.3	5.7	12.6	20.0	8.5	6.0	7.1	6.3	30.3	11.4	10.3	(16)
	PrecMin	26.6	0.0	0.0	0.0	1.2	0.8	0.6	0.6	1.1	0.1	0.0	0.0	0.0	(17)
	PrecStd	14.3	1.9	0.9	1.5	3.5	4.5	2.1	1.3	1.3	1.7	6.0	2.8	2.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.9	93.0	93.4	91.8	89.6	86.3	85.5	84.6	86.2	88.0	89.8	91.7	(19)
		MCWB	76.6	77.0	77.9	78.0	75.6	74.1	73.2	72.5	73.5	75.7	77.9	78.0	(20)
	2%	DB	91.0	91.4	91.7	91.0	87.8	85.1	84.0	84.0	84.6	86.3	88.1	90.0	(21)
		MCWB	76.6	77.0	77.8	77.8	76.2	73.8	72.4	72.3	73.1	75.4	77.6	77.6	(22)
	5%	DB	89.7	90.0	91.0	89.6	86.2	84.2	82.6	82.6	84.1	85.8	87.6	89.4	(23)
		MCWB	76.4	76.7	77.7	77.6	75.9	73.6	72.2	72.1	73.1	75.3	77.5	77.6	(24)
	10%	DB	88.4	89.4	89.7	88.0	84.7	82.8	82.0	82.0	82.7	84.4	86.2	88.0	(25)
		MCWB	76.3	76.5	77.5	77.6	75.8	73.6	72.2	72.1	72.8	74.9	77.1	77.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.2	79.8	80.7	80.1	78.9	77.2	75.4	75.4	76.4	78.4	79.9	80.6	(27)
		MCDB	86.9	88.3	87.7	86.4	84.5	82.2	80.5	80.0	81.6	83.7	86.2	87.7	(28)
	2%	WB	78.3	78.8	79.8	79.4	78.1	76.2	74.5	74.4	75.3	77.5	79.0	79.5	(29)
		MCDB	86.0	86.8	86.5	85.6	83.5	81.4	79.9	79.4	80.6	82.6	84.9	86.1	(30)
	5%	WB	77.7	78.1	79.2	79.1	77.4	75.5	73.8	73.7	74.6	76.8	78.3	79.0	(31)
		MCDB	85.4	85.8	85.7	85.1	82.7	80.7	79.1	78.9	79.9	81.7	84.0	85.4	(32)
	10%	WB	77.1	77.4	78.9	78.7	76.9	74.8	73.2	73.0	74.2	76.2	78.0	78.4	(33)
		MCDB	84.8	84.9	85.3	84.6	82.1	80.0	78.5	78.3	79.5	81.2	83.5	84.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	14.3	13.3	11.8	10.7	11.3	11.8	12.0	11.9	11.4	11.6	13.1	(35)
		MCDBR	15.0	14.8	14.3	13.1	12.5	12.5	12.8	12.9	12.9	12.4	12.7	14.0	(36)
	5% WB	MCWBR	4.1	4.0	4.1	4.0	4.5	4.7	4.6	4.6	4.3	4.1	4.3	4.4	(37)
		MCWBR	14.2	14.2	13.1	12.2	11.0	11.1	11.7	11.7	11.6	10.9	11.8	13.0	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.482	0.490	0.450	0.411	0.422	0.442	0.441	0.425	0.414	0.415	0.416	0.460	(40)
		taud	2.222	2.193	2.355	2.497	2.445	2.343	2.325	2.374	2.400	2.419	2.440	2.317	(41)
	Ebn at Noon	Ebn at Noon	272	271	279	282	270	260	263	274	285	288	290	278	(42)
		Edh at Noon	48	49	42	35	35	38	39	39	39	39	38	43	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1896	1978	1932	1711	1493	1470	1512	1630	1827	1904	1910	1898	(44)
	RadStd	RadStd	91	71	104	103	125	69	69	69	82	89	98	80	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	+0.65	+0.85	+0.83	+0.56	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page